

TailGuard Diagnostics for Activation-Aware PTQ

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Abstract

Activation-aware post-training quantization (PTQ) often relies on the premise that a small number of activation channels deserve special treatment, but a plausible channel rule is not itself evidence of downstream quality. We study TailGuard as a calibration-only diagnostic that ranks activation channels, protects a small heavy-tail fraction, and routes the remaining activations through an INT4 path. On 128-example WikiText-2, MMLU, and GSM8K slices, we compare the default $p = 0.015$ TailGuard row with FP16 references, uniform activation-hook controls, and AWQ W4A16. The evidence yields a negative diagnostic result: AWQ is the strongest usable low-bit PTQ comparator, with WikiText-2 mean negative loss -4.49091280159 , MMLU accuracy 0.5234375 , and GSM8K accuracy 0.8515625 , while default TailGuard is worse than AWQ and FP16 on the usable families (-9.46083122492 on WikiText-2 and 0.2421875 on MMLU). The one-row MMLU AWQ-FP16 edge is audited as 67 versus 66 correct, with 11 AWQ-only and 10 FP16-only wins, so we treat it only as a slice/scoring artifact candidate rather than a quantization-side quality gain. GSM8K has no usable TailGuard quality row because the activation-hook rows are excluded as no-go provenance, so the paper contributes a scoped negative diagnostic: it establishes AWQ as the required comparator, separates usable negative rows from invalid execution paths, and states the repairs needed before any future TailGuard method claim.

1 Introduction

Low-bit PTQ is attractive because it can reduce model memory without retraining, but strong PTQ evidence depends on more than a plausible compression mechanism. GPTQ shows that weight-only quantization can use calibration and second-order information to preserve transformer quality after training [2]. AWQ protects salient weights under activation-aware calibration and gives this paper a practical published comparator for LLM compression [4].

A separate line of work motivates activation-side diagnostics. LLM.int8() routes activation outliers through higher precision, showing that a small set of features can matter disproportionately for low-bit inference [1]. SmoothQuant shifts

activation difficulty into weights through calibration-derived scaling, again treating activation structure as a source of quantization difficulty [3]. Other PTQ systems use calibration or transformations to change the low-bit problem before inference: OmniQuant calibrates quantization parameters, while QuaRot and SpinQuant use rotations to reduce outlier pressure [5, 8, 9]. The open question for a cheap activation-channel rule is therefore not whether activation tails exist, but whether protecting a small ranked subset yields scoreable downstream quality against a mature PTQ baseline.

TailGuard tests that narrower question. It uses a train-free calibration pass to rank activation channels, protects a heavy-tail fraction, and sends the remaining activations through an INT4 path. This design is intentionally lighter than learned rotations or task-label tuning, which makes it useful as a diagnostic probe. The same simplicity also raises the evidentiary bar: a small movement over a uniform activation-hook control is not enough if AWQ remains stronger, and an invalid generation path cannot be converted into an ordinary accuracy number. The evaluation therefore treats row denominators, paired identifiers, and row-validity decisions as part of the scientific result.

The measured answer is unfavorable for TailGuard. On WikiText-2, default TailGuard scores -9.46083122492 , far below AWQ at -4.49091280159 and FP16 at -4.39036326762 under the reported mean-negative-loss convention. On MMLU, default TailGuard scores 0.2421875 , below AWQ at 0.5234375 and FP16 at 0.515625 . The AWQ-FP16 MMLU gap is only one item, and the item audit records 56 both-correct rows, 51 both-wrong rows, 11 AWQ-only wins, and 10 FP16-only wins; the paper therefore does not interpret that slice as evidence that quantization improves MMLU quality over FP16. On GSM8K, the scoreable rows are FP16 at 0.890625 and AWQ at 0.8515625 ; the activation-hook TailGuard row is excluded because the no-go slice collapses to constant-zero scored rows and cannot support an ordinary accuracy estimate. The result is not a near miss hidden by presentation. It is a measured pattern of TailGuard underperformance against the AWQ comparator and FP16 reference on scoreable rows, plus execution invalidity on the missing reasoning row.

This paper is consequently framed as a negative diagnostic study with three contributions. First, it states a concrete TailGuard protocol and its fixed effective-bit accounting so

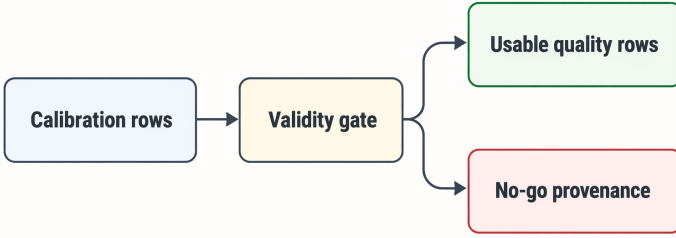


Figure 1: Evidence-routing schematic. Candidate rows pass through a validity gate before the manuscript separates retained quality evidence from no-go provenance.

the evaluated method is not moved after the result. Second, it reports a 128-example, three-family diagnostic matrix that places AWQ beside TailGuard wherever both rows are usable. Third, it gives a failure analysis that separates usable negative rows from invalid activation-hook execution paths. The claim is not that TailGuard works. The claim is that the measured evidence says where this default path fails and what must be repaired before a competitive PTQ method claim would be justified.

Figure 1 gives the paper-level routing view: active evidence is filtered into usable AWQ-anchored rows and excluded no-go provenance, rather than merged into a single positive TailGuard score.

2 Related Work

Weight-only PTQ. GPTQ uses second-order information to quantize transformer weights after training [2]. AWQ uses activation-aware calibration to protect salient weights and is a practical strong baseline for LLM compression [4]. This study keeps AWQ in the main result table because it prevents the evaluation from comparing TailGuard only against a weak uniform activation-hook control.

Activation outliers. LLM.int8() routes outlier features through higher precision to preserve matrix multiplication quality [1]. SmoothQuant migrates activation difficulty into weights with calibration-derived scaling [3]. TailGuard shares the premise that activation tails matter, but the present rows do not show that a simple protected-channel mask is competitive with mature PTQ baselines.

Transformations and rotations. OmniQuant calibrates quantization parameters across low-bit settings [5]. QuaRot and SpinQuant use rotations to reduce outlier pressure before low-bit inference [8, 9]. TailGuard deliberately avoids learned rotations, adapters, or task-label optimization. That boundary makes the diagnostic cheap, but the observed evidence shows that cheapness alone is not enough for downstream quality.

Adjacent quantization settings. KVQuant and KIVI study cache compression under long-context constraints [6, 7]. Q-VLM studies PTQ for large vision-language models [10]. These directions motivate broader activation-tail questions, but this diagnostic matrix contains no multimodal row and no

multimodal performance claim.

3 Method

TailGuard has four stages. First, unlabeled calibration prompts are passed through the model while activation hooks collect per-layer channel statistics. Second, each channel receives a rank score. The default row in the diagnostic matrix uses a heavy-tail score with protected fraction $p = 0.015$. Third, the highest-ranked channels are marked as protected. Fourth, unprotected activations use an INT4 path while protected channels remain in a higher-precision side stream.

The precision accounting is:

$$B_{\text{eff}} = (1 - p) \cdot 4 + p \cdot 16. \quad (1)$$

For the default protected fraction $p = 0.015$, this gives an effective activation budget of 4.18 bits. This value is an accounting convention, not a hardware-equivalent speed claim.

The method is train-free in this study. It uses no fine-tuning, learned adapters, learned rotations, task-label optimization, simulator override, or post-hoc answer repair. This boundary is important because the negative result should not be reinterpreted as a failure of a trained quantizer. It is a failure of the specific calibration-only activation-protection path measured here.

4 Experimental Setup

The diagnostic matrix contains 10 included 128-example quality rows and two excluded activation-hook no-go rows across WikiText-2, MMLU, and GSM8K. No table or claim in the main manuscript averages excluded activation-hook rows into ordinary quality estimates.

Benchmarks and models. WikiText-2 uses the Salesforce WikiText test split and Qwen/Qwen2.5-1.5B. MMLU uses the cais/mmlu test split and Qwen/Qwen2.5-7B-Instruct. GSM8K uses the openai/gsm8k test split and Qwen/Qwen2.5-7B-Instruct. Each included aggregate condition reports 128 scored rows. The public splits are larger than the evaluated slices, so the paper calls the evidence a diagnostic slice rather than full public-split evidence.

Baselines. FP16 is the quality reference. Uniform INT4 activation hook is the closest implementation control for TailGuard on WikiText-2 and MMLU. AWQ W4A16 is the strongest usable PTQ baseline in the evaluated matrix. GSM8K retains only FP16 and AWQ quality rows; the activation-hook uniform INT4 and TailGuard rows are excluded from quality claims.

Metrics and uncertainty. WikiText-2 reports mean negative loss, where higher is better by definition of the reported score. MMLU and GSM8K report accuracy. Binary rows include Wilson intervals in the canonical result table, and paired uncertainty is computed over matched example identifiers. Comparative language remains descriptive except where the paired uncertainty artifact directly supports a direction for the same slice.

Exclusion policy. An included row must produce answer-bearing outputs that can be scored under the declared metric. GSM8K activation-hook uniform INT4 and TailGuard rows are retained as excluded-row evidence because their score distributions are constant zero and the generation path did not pass the validity checks required for ordinary quality estimates. They are not converted into ordinary accuracy estimates, and they are not hidden as missing data.

Row comparability. The main comparisons are organized by benchmark family rather than averaged into one score. WikiText-2 compares language-modeling loss rows under the same model family and metric. MMLU compares multiple-choice accuracy rows under the same denominator. GSM8K compares only the rows that passed the answer-bearing generation and scoring checks. This prevents a single aggregate from mixing loss, multiple-choice accuracy, reasoning exact match, usable rows, and excluded execution failures.

Paired protocol. Paired comparisons use shared example identifiers within a benchmark family, not a cross-benchmark aggregate. For WikiText-2, the paired delta is the per-example change in reported mean negative loss under the higher-is-better convention. For MMLU and GSM8K, the paired delta is the candidate’s correctness indicator minus the baseline’s correctness indicator on the same question. A paired row is reported only when both conditions pass the answer-bearing validity rule. This is why the GSM8K TailGuard comparison has no ordinary paired accuracy delta: the TailGuard activation-hook execution path is retained as no-go provenance, not as a quality row.

Validity denominator. The denominator in the main table is therefore not only a sample-size count. It is also a validity statement. A 128-example entry means that the condition produced 128 scoreable examples under the declared metric. The excluded GSM8K activation-hook rows have smaller observed no-go slices and constant-zero scored outcomes, so they are routed to failure analysis rather than inflated into the 128-example quality table. This rule is conservative for both sides of the comparison: it prevents invalid TailGuard generations from being hidden, and it prevents invalid execution collapse from being described as ordinary low task accuracy.

The same denominator rule also determines how uncertainty is read. A paired interval is meaningful only after both rows survive the validity gate and share example identifiers. Thus the WikiText-2 and MMLU TailGuard–AWQ comparisons are paired diagnostic comparisons, while the GSM8K TailGuard–AWQ cell is a validity failure rather than a wide confidence interval. This distinction is important for a negative paper: it prevents the strongest result from being overstated as a universal PTQ failure, and it prevents an invalid reasoning path from being treated as if it were merely a noisy sample.

The protocol deliberately separates three notions that are often conflated in small PTQ studies. A *quality row* is an included row with scoreable outputs under the declared metric. A *mechanism control* tests the nearest activation-hook implementation path, but only when that path itself produces valid outputs. A *no-go row* is preserved evidence for a failed execution path and cannot be averaged into quality estimates.

Calibration and masking therefore define only the measured activation path; the validity gate still decides whether a scored row can support a quality claim. TailGuard has usable negative quality rows on WikiText-2 and MMLU; it has no usable GSM8K quality row in this diagnostic matrix.

5 Results

Table 1 is the central result. It keeps the body view to the comparison-critical fields while also exposing source, split scale, and evaluated model. AWQ is shown beside TailGuard wherever both rows are usable; GSM8K reports only the scoreable FP16 and AWQ quality rows. Appendix A records paired-row provenance and excluded no-go details that would otherwise make the main table read like an execution log.

The result blocks a positive method claim. On WikiText-2, default TailGuard is only slightly above the uniform INT4 activation-hook control and remains roughly five mean-negative-loss points behind AWQ and FP16. On MMLU, TailGuard is below AWQ and FP16 and only marginally above uniform INT4; AWQ’s one-row edge over FP16 is shown for auditability, not claimed as a quantization-side gain. On GSM8K, there is no usable TailGuard quality row; the only retained PTQ quality row is AWQ at 0.852 against FP16 at 0.891 in the rounded body table.

The WikiText-2 block is the language-modeling stress test. All four rows share the denominator and score direction, so the slice directly shows that the default TailGuard row remains close to the uniform activation-hook control and far from AWQ and FP16. The protected-channel path may measure a relevant activation property, but this recipe does not preserve language-modeling quality at the level of the published PTQ baseline.

The MMLU block is negative evidence for downstream transfer. TailGuard is slightly above the nearest activation-hook control, but Table 1 shows it far below AWQ and FP16 on the same denominator. This is not a formatting artifact or a missing-baseline problem; the default protected-channel recipe does not transfer to this multiple-choice slice at a level relevant to the PTQ comparison.

The GSM8K block separates task quality from execution validity. FP16 and AWQ produce scoreable answer-bearing rows, while the activation-hook TailGuard and uniform INT4 rows are excluded because their scored rows collapse to constant zero. The paper therefore does not present a TailGuard GSM8K accuracy number: the measured activation-hook path is not yet a valid GSM8K quality path.

The negative result has different strength in each family: TailGuard *underperforms* on WikiText-2 and MMLU, the GSM8K activation-hook path *fails validity*, and AWQ *anchors* the strongest usable PTQ comparison. A family-specific table avoids mixing loss, accuracy, usable rows, and invalid execution paths into one misleading macro-average.

Table 1: Diagnostic 128-example matrix: AWQ anchors the usable PTQ comparison, while default TailGuard heavy-tail $p=0.015$ is weak or negative. Higher is better for both metrics; GSM8K TailGuard is excluded from ordinary quality rows, and the one-row MMLU AWQ-FP16 edge is not interpreted as a quantization-side quality gain.

Family	Source / model	Method	Budget	Metric	Score
WikiText-2	Salesforce 128/4358; Qwen2.5-1.5B	FP16 reference	16w/16a	neg. loss	-4.390
WikiText-2	Salesforce 128/4358; Qwen2.5-1.5B	uniform INT4 control	16w/4a	neg. loss	-9.483
WikiText-2	Salesforce 128/4358; Qwen2.5-1.5B	TailGuard $p=0.015$ negative	16w/4.18a	neg. loss	-9.461
WikiText-2	Salesforce 128/4358; Qwen2.5-1.5B	AWQ W4A16 anchor	4w/16a	neg. loss	-4.491
MMLU	cais 128/14042; Qwen2.5-7B-Instruct	FP16 reference	16w/16a	accuracy	0.516
MMLU	cais 128/14042; Qwen2.5-7B-Instruct	uniform INT4 control	16w/4a	accuracy	0.219
MMLU	cais 128/14042; Qwen2.5-7B-Instruct	TailGuard $p=0.015$ negative	16w/4.18a	accuracy	0.242
MMLU	cais 128/14042; Qwen2.5-7B-Instruct	AWQ W4A16 anchor	4w/16a	accuracy	0.523
GSM8K	openai 128/1319; Qwen2.5-7B-Instruct	FP16 reference	16w/16a	accuracy	0.891
GSM8K	openai 128/1319; Qwen2.5-7B-Instruct	AWQ W4A16 anchor	4w/16a	accuracy	0.852

Note. Each source/model cell reports the evaluated split count and model/backend. All listed rows have $n = 128$ scoreable examples. Additional paired-row and excluded-row provenance appears in Appendix A.

6 Diagnostic Analysis and Ablation

The diagnostic evidence supports paired uncertainty for the main usable comparisons and records why GSM8K TailGuard comparisons are not testable. TailGuard-AWQ is directionally negative on WikiText-2 and MMLU; GSM8K TailGuard-AWQ is absent because TailGuard is excluded no-go provenance. The canonical paired-uncertainty artifact retains the matched-row details.

The paired analysis strengthens the negative boundary without creating a broad significance claim. WikiText-2 and MMLU show the same pattern: TailGuard is near the uniform activation-hook control and far below AWQ and FP16. These rows reject the current method story for this diagnostic slice; they do not rank all PTQ methods for the benchmark.

The evaluated matrix does not contain a fresh rank-metric or protected-fraction ablation grid. Earlier exploratory rows can motivate future controls, but the manuscript does not report them as a recovered recipe. A weak ablation lead on one slice would not change the core result: the default measured TailGuard path trails AWQ and FP16 on the usable rows and lacks a valid GSM8K quality row.

What the diagnostics can say. The evidence can say that AWQ is the strongest usable PTQ baseline in the evaluated matrix; that default TailGuard is weak or negative on WikiText-2 and MMLU; and that GSM8K activation-hook rows are no-go provenance, not quality estimates. It can also say that the paired rows point in the negative direction for TailGuard versus AWQ on the usable families.

MMLU AWQ-FP16 audit. The MMLU AWQ-FP16

paired slice has $n = 128$, with 67 AWQ-correct rows and 66 FP16-correct rows. The disagreement pattern is nearly balanced: 56 rows are correct under both methods, 51 are wrong under both methods, 11 are AWQ-only wins, and 10 are FP16-only wins. The paired-uncertainty artifact marks the one-row delta as not statistically decisive, and the item audit records the edge as a slice/scoring-artifact candidate. The manuscript therefore uses the row only to bound interpretation; it does not claim that AWQ improves MMLU quality over FP16.

What the diagnostics cannot say. The evidence cannot claim a robust TailGuard mechanism, an optimal protected fraction, a recovered rank score, a GSM8K TailGuard quality value, an AWQ-over-FP16 quantization gain, or a multimodal result. It also cannot claim public-split performance. The paper keeps those absences visible because hiding them would turn a useful negative diagnostic into an unsupported method paper.

Next-measurement order. The evidence determines what should be measured next. First, GSM8K activation-hook generation must be made answer-bearing under the same prompt and stopping rule. Second, rank-score and protected-fraction ablations should be rerun with random-mask controls. Third, larger matched slices should test sampling stability only after those mechanism and validity questions are separated. Running a larger copy of the current default recipe would not answer the causal question.

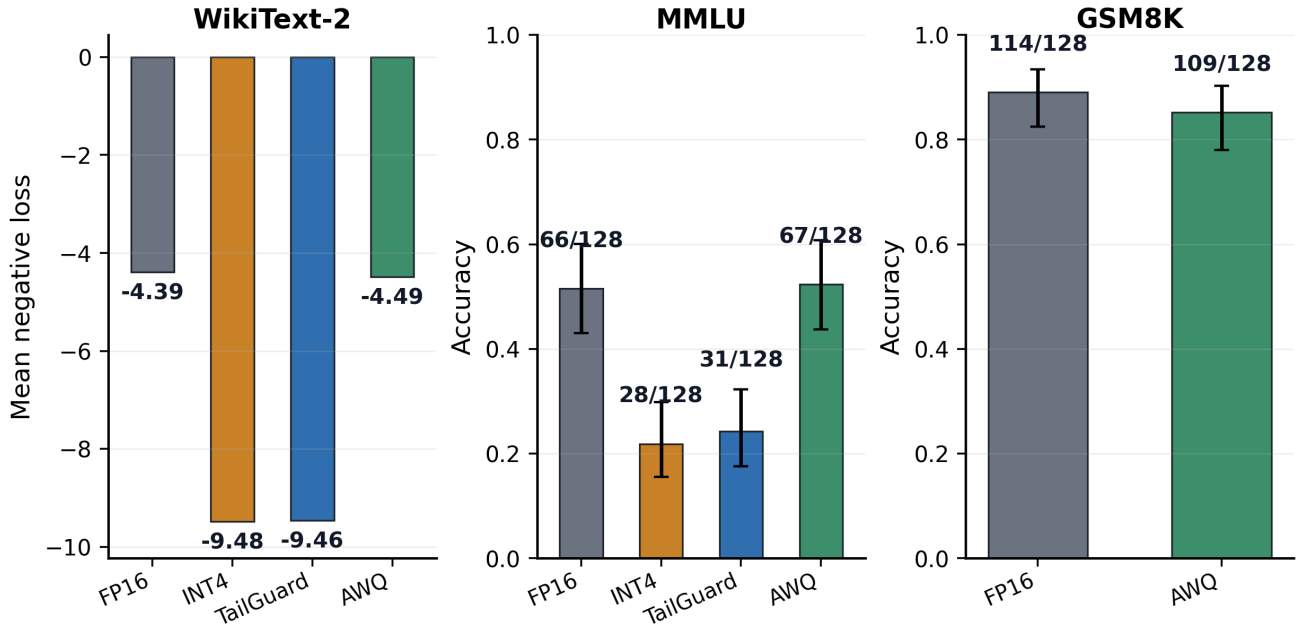


Figure 2: Diagnostic result summary from the 128-example matrix. TailGuard does not beat AWQ, and GSM8K has no usable TailGuard activation-hook quality row.

7 Failure Cases

The main failure case is GSM8K activation-hook collapse. The excluded TailGuard $p = 0.015$ activation-hook row has 8 scored rows, all score 0.0, and the related uniform INT4 activation-hook row has 22 scored rows with one unique zero score. Both rows are preserved as no-go provenance, not as ordinary accuracy estimates, because the measured generation path did not produce answer-bearing reasoning outputs under the declared rule. Figure 4 summarizes the row-accounting boundary, and Appendix A gives the full provenance details.

This distinction is strict in both directions. If an excluded GSM8K activation row were hidden, the manuscript would overstate TailGuard by omission. If it were reported as ordinary zero accuracy, the manuscript would understate the task construct by mixing a broken execution path with answer-level model quality. The bounded diagnosis is narrower: the measured activation-hook reasoning path is not yet a valid quality-measurement path.

The no-go rows also explain why the paper does not treat every scored record as equivalent evidence. A scalar score can be present even when the generation path has failed the scientific construct. In the GSM8K activation-hook slices, the visible symptom is a constant-zero distribution, but the deeper problem is that the generations do not provide the answer-bearing completions that the math-reasoning metric assumes.

Two additional failures keep the result diagnostic rather than prescriptive. First, the evidence lacks random protected masks, fresh rank-metric ablations, calibration-domain splits, layer-family comparisons, and protected-fraction sweeps under the same matched protocol. Without those controls, the paper can identify an observed row difference but cannot claim

that the heavy-tail score caused a transferable improvement. Second, each included condition has 128 scored rows. That scale catches catastrophic failures and gives a useful diagnostic benchmark, but it is not a full public benchmark estimate.

Baseline dominance. AWQ dominates the usable low-bit PTQ comparison against TailGuard in the evaluated matrix. On WikiText-2 it is close to FP16 and far above default TailGuard. On MMLU it is far above TailGuard, while its one-item numeric edge over FP16 is treated as an audited slice artifact candidate rather than a quality-gain claim. On GSM8K it is the only usable PTQ row. Baseline dominance is not a formatting issue; it is the central scientific outcome for the TailGuard comparison.

Non-transfer. TailGuard’s small WikiText-2 movement over uniform INT4 does not translate into a useful downstream win on MMLU. A PTQ method must preserve downstream task behavior, not only move a loss slice relative to a local activation-hook control. The present data do not isolate whether the non-transfer comes from the rank score, protected fraction, activation-hook implementation, or sample noise, but they do reject the broad statement that default TailGuard improves quality across the measured families.

Execution-path invalidity. The GSM8K activation-hook rows show an invalid execution path rather than ordinary wrong answers. In a deployment setting, this is severe: a quantized path that does not produce answer-bearing completions cannot be rescued by a better scorer. The paper therefore treats execution validity as a first-class result alongside quality.

8 Validity Routing and Next Measurements

The row-validity rule is part of the result. Included rows support descriptive quality statements; excluded GSM8K

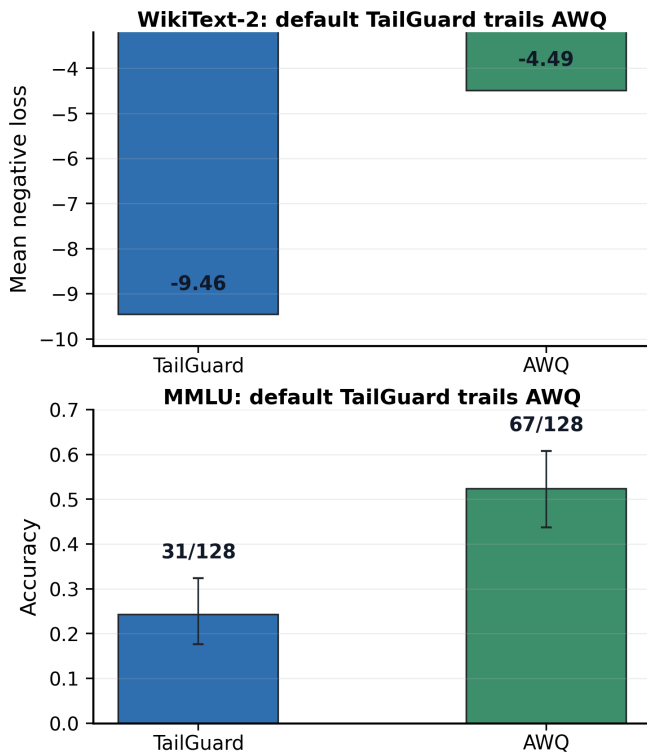


Figure 3: Method-slice diagnostic view. The evaluated rows show that the default TailGuard activation-protection path remains far from AWQ on the usable families.

activation-hook rows support only validity statements; missing multimodal rows support no result claim. This routing keeps invalid execution paths visible without promoting invalid generations into ordinary accuracy estimates.

Together, Table 1 and Appendix A define the decision boundary. AWQ is interpreted as a usable PTQ row wherever it passes the same accounting rule; uniform INT4 activation-hook rows are mechanism controls only when their generations remain valid; and TailGuard rows are split between usable negative evidence on WikiText-2/MMLU and excluded execution-path evidence on GSM8K. The three families are not averaged into a single score: WikiText-2 reports mean negative loss, MMLU reports multiple-choice accuracy, and GSM8K requires answer extraction before exact-match scoring. TailGuard’s 4.18-bit activation entry is likewise an accounting field, not a hardware-normalized latency or memory claim.

The paired evidence has the same boundary. WikiText-2 and MMLU have matched TailGuard–AWQ rows, so their paired artifacts support directional negative statements for this diagnostic slice. GSM8K has no valid TailGuard quality row, so the TailGuard–AWQ comparison is absent rather than merely imprecise. A repaired study should first produce answer-bearing GSM8K activation-hook generations, then preserve matched identifiers against FP16 and AWQ, and then test whether the protected-channel ranking beats same-budget random or alternate-score masks. Until those rows change, the fixed TailGuard recipe should not be relabeled as a method

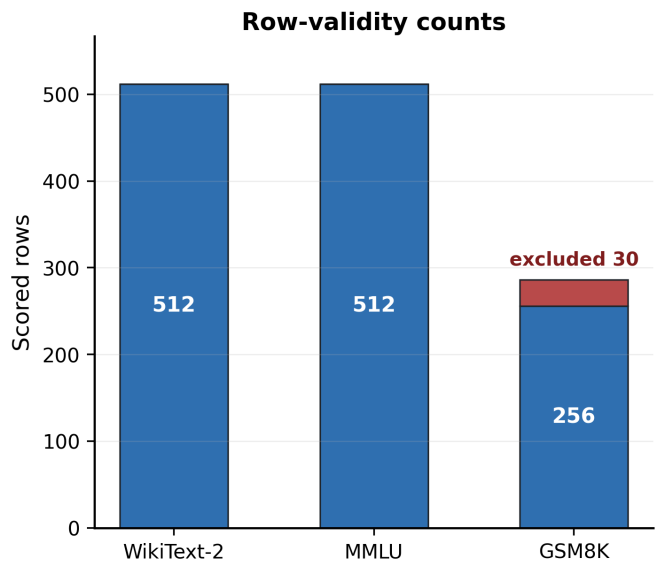


Figure 4: Row-validity accounting. Included rows support descriptive quality claims; excluded GSM8K activation-hook rows support only no-go provenance and method-failure diagnosis.

win.

Scalar scores are not enough. The validity rule is stricter than checking whether a scalar score exists. A row can emit a numeric value while failing the answer-bearing construct required by the task metric. That is why the GSM8K activation-hook rows remain visible but excluded: they record a real execution failure, yet they cannot support an ordinary exact-match quality estimate. This treatment is conservative in both directions. It prevents a failed TailGuard reasoning path from being hidden as missing data, and it also prevents that path from being described as if a valid model simply answered every example incorrectly.

Family-local interpretation. The diagnostic matrix can be rebuilt from three row groups. WikiText-2 contains FP16, uniform INT4, TailGuard, and AWQ under the same language-modeling metric. MMLU contains the same four method roles under the same multiple-choice denominator. GSM8K contains only FP16 and AWQ as included quality rows, while the activation-hook rows are retained as excluded evidence. These groups answer related but different measurement questions: loss preservation, multiple-choice transfer, and answer-bearing reasoning validity.

Comparator discipline. AWQ is included because it is the strongest usable PTQ comparator in the evaluated matrix, not because this paper makes a broad AWQ benchmark claim. On the scoreable families, AWQ anchors the practical bar that TailGuard would need to clear. The comparison keeps the diagnostic honest: a protected-channel recipe is not competitive merely because it moves slightly above a weak local activation-hook control on one slice. FP16 remains the quality reference, so the paper also avoids treating the one-row MMLU AWQ–FP16 edge as a quantization-side gain.

No-go rows are evidence, not table clutter. Excluded rows are retained because a negative diagnostic should show

both kinds of failure: weak scoreable rows and invalid execution paths. Hiding the GSM8K activation-hook rows would make the evidence look cleaner than it is. Folding them into Table 1 as ordinary zero-accuracy rows would make the task construct look worse than it is. The retained no-go category preserves the observation without overstating its metric meaning.

Reuse boundary. The reusable result is not that activation tails are irrelevant, that every protected-channel method fails, or that AWQ is globally optimal. The reusable result is narrower: keep AWQ and FP16 in the comparison set, preserve paired identifiers, route invalid generations separately, and require the GSM8K path to become answer-bearing before quoting TailGuard accuracy. This makes the negative claim falsifiable by new rows rather than by new wording.

Follow-up reporting rule. Future rows should be reported in the same order as this paper’s decision rule: first state whether the generation path is answer-bearing, then state the denominator, then report paired comparisons against FP16 and AWQ, and only then summarize any aggregate. This order prevents a repaired run from inheriting the present negative claim, but it also prevents an unrepaired run from being promoted by a more favorable table layout.

Denominator audit. The row denominator is part of the evidence claim. For included rows, $n = 128$ means 128 scoreable examples under the stated metric. For excluded rows, the smaller observed no-go slice is not promoted to the quality denominator. Reporting those denominators before interpretation is what keeps the negative result auditable and prevents a failed execution path from being hidden inside a cleaner-looking aggregate.

9 Discussion and Limitations

The claim scope is intentionally narrow. The 128-example diagnostic slices show that default TailGuard underperforms AWQ and FP16 on the scoreable WikiText-2 and MMLU rows, while the GSM8K activation-hook rows support only execution-path statements. The paper therefore avoids a single headline score, SOTA language, broad compression claims, multimodal or long-context claims, and hardware-normalized deployment claims.

The next useful TailGuard study should change the evidence before changing the claim. It needs answer-bearing GSM8K activation-hook generations, matched rows against FP16 and AWQ, and same-budget random-mask or alternate-score controls under the same row-validity rule. Larger public-split slices are useful only after those validity and mechanism controls are in place.

The 128-example scale also fixes the strength of the conclusion. It is enough to expose the default recipe’s weak scoreable rows and the GSM8K execution failure, and it is enough to require AWQ in future comparisons. It is not enough to choose a final protected fraction, rank score, or deployment policy. Those decisions require a repaired mechanism, paired public-split expansion, and hardware measurements that this diagnostic deliberately does not claim.

The value of the negative result is the separation of failure modes. A weak scoreable row calls for a stronger mechanism or comparator-aware ablation. An invalid activation-hook row calls for an execution repair before accuracy can be discussed at all. Treating those outcomes separately keeps future work from optimizing the wrong surface: it should not tune wording around a collapsed reasoning path, and it should not evaluate a new rank score without the AWQ baseline and paired identifiers that make the present diagnostic interpretable.

10 Conclusion

This diagnostic rejects a positive claim for the fixed calibration-only TailGuard row. The measured contribution is the boundary itself: AWQ remains the required usable comparator, default TailGuard is negative on the scoreable families, and GSM8K activation-hook evidence must remain no-go provenance until it produces answer-bearing completions under the declared rule. A positive follow-up needs answer-bearing GSM8K rows plus matched FP16, AWQ, activation-hook-control, and same-budget mask comparisons.

The paper therefore leaves a falsifiable checklist rather than a tuned success story. A stronger TailGuard claim would need new included rows that change the comparison, not a different reading of the present rows: the reasoning path must pass the same validity gate, the protected-channel rule must beat both AWQ and same-budget controls on shared identifiers, and the one-row AWQ–FP16 MMLU edge must remain an audit caveat rather than a mechanism claim. Until those measurements exist, the negative diagnostic is the submission-ready interpretation of this evidence bundle.

That interpretation is useful precisely because it separates design failure from measurement failure. The design side is visible in the scoreable WikiText-2 and MMLU rows, where the default protected-channel rule remains far below the AWQ comparator. The measurement side is visible in GSM8K, where the activation-hook path must first produce answer-bearing completions before an accuracy row can be meaningful. A future paper can overturn either part with new rows, but it should not combine them into a single optimistic aggregate. Keeping those two failures separate is the main lesson for activation-aware PTQ evaluation under small diagnostic budgets.

For review purposes, this means the paper should be judged on whether the boundary is useful and reproducible, not on whether TailGuard wins. The current answer is reproducible within the retained artifacts: AWQ is the comparator to beat, the default TailGuard row does not beat it on scoreable slices, and the GSM8K path is a validity failure rather than a hidden accuracy value. That is a complete negative diagnostic result, with the next measurements specified before any stronger method claim is allowed. The main contribution is therefore a calibrated stop condition for this recipe and a concrete measurement contract for the next one.

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Reproducibility Checklist

Scope and evidence. All reported manuscript numbers come from scored rows linked to public benchmark slices and re-

tained companion artifacts. The paper reports completed diagnostic slices rather than full public-split estimates.

Claims. The manuscript makes no SOTA, broad significance, multimodal, TailGuard GSM8K quality, or TailGuard method win claim. The reported claims are negative or diagnostic and match the measured rows.

Artifacts. The result tables, scored-row files, raw model outputs, paired alignment, and row-validity records are retained as companion artifacts. The manuscript keeps only reader-facing labels and numbers in rendered prose.

Figures. Data figures are generated from the same numeric artifacts as the tables. Conceptual figures illustrate the method and diagnostic framing; they do not introduce additional quantitative evidence.

Result reconstruction. The main table can be reconstructed by grouping rows by benchmark family, method, metric, and inclusion decision. WikiText-2 uses the mean of 128 row losses under the reported sign convention. MMLU and GSM8K use the number of correct examples divided by 128 for included rows. The excluded GSM8K activation-hook rows are not averaged into those quality rows; they are reported only through the row-validity analysis because they fail the answer-bearing generation rule.

A Appendix: Evidence Routing

The paper intentionally routes rows by validity category. Included rows support descriptive quality statements. Excluded GSM8K activation-hook rows support only excluded-row validity statements. Missing multimodal rows support no result claim. This routing is the binding rule for future evaluations: stronger language requires new included rows, not reinterpretation of excluded or absent evidence.

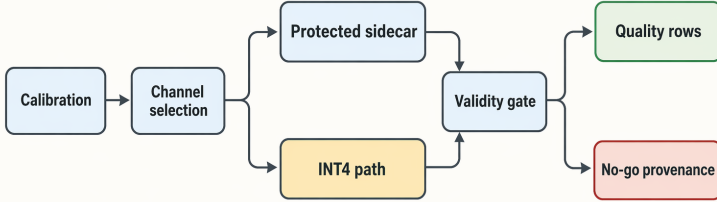


Figure 5: Appendix method/pipeline schematic. The diagnostic path selects protected activation channels, routes the remaining activations through the low-bit path, and then applies the same validity gate that separates retained quality rows from excluded no-go provenance.

Table 2: Benchmark provenance and included quality rows for the 128-example diagnostic matrix.

Family	Slice/model	Included quality rows
WikiText-2	Salesforce test, 128/4358; Qwen2.5-1.5B	FP16, INT4, TailGuard, AWQ
MMLU	cais test, 128/14042; Qwen2.5-7B-Instruct	FP16, INT4, TailGuard, AWQ
GSM8K	openai test, 128/1319; Qwen2.5-7B-Instruct	FP16 and AWQ only

Table 2 keeps benchmark provenance and row-validity interpretation together so the appendix can be read without decoding compressed audit cells. Uniform INT4 denotes the activation-hook control, and TailGuard denotes the default $p = 0.015$ row.

GSM8K no-go provenance. Two activation-hook rows are retained as excluded evidence rather than quality rows. The uniform INT4 hook is a mechanism-control row with 22 scored rows, one unique score value, null parse symptoms, capped generations, and an excluded no-go decision. The TailGuard $p = 0.015$ activation-hook row is a method-path row with 8 scored rows, one unique score value, the same null-parse/capped-generation symptoms, and the same excluded no-go decision.

This routing also defines how negative rows should be cited. A complete TailGuard MMLU row with 128 scored examples is a usable negative result. It can support a statement that the measured method underperforms FP16 and AWQ on that slice. An invalid GSM8K activation-hook row cannot support a quality estimate, even if its scored rows are all zero.

It can support only the statement that the measured execution path failed the validity rule.

Paired/provenance companion. The paired artifact keeps comparison status out of the table cells. WikiText-2 and MMLU TailGuard-AWQ both have paired $n = 128$ and negative deltas for TailGuard. MMLU AWQ-FP16 has paired $n = 128$ with delta 0.0078125 and is not decisive; the row-level audit resolves that as 67 versus 66 correct with 11 AWQ-only and 10 FP16-only wins. GSM8K TailGuard-AWQ has paired $n = 0$ because TailGuard is excluded no-go provenance; GSM8K AWQ-FP16 has paired $n = 128$ with delta -0.0390625 , which is not decisive.

B Appendix: Follow-Up Experiment Requirements

The next TailGuard evaluation should keep shared example identifiers across FP16, uniform INT4, TailGuard, and AWQ wherever paired interpretation is available. It should first repair the GSM8K activation-hook path and verify complete numeric answers under the declared extraction rule. It should then run matched MMLU and WikiText-2 comparisons that vary the rank score, random masks, and protected fraction while recording the effective-bit budget beside each row. Those measurements should be completed before any positive TailGuard claim is drafted.